RUUD PEREZ

Industrial Engineering | Simulation, Operations & Supply Chain Analytics
NYC Metro Area | ruud.perez@gmail.com | 201-761-9192 | ruud0.github.io | github.com/ruud0

PROFESSIONAL SUMMARY

Industrial Engineering student (May 2027) who builds simulation and capacity models to test operating strategies. Best result so far: showing a scheduler that cut vehicle stops about fivefold also more than doubled pedestrian wait.

EDUCATION

The Pennsylvania State University | B.S. Industrial Engineering | Expected May 2027
Relevant coursework: Operations Research, Stochastic Operations Research, Service Systems Engineering (supply chain), Statistical Methods, Industrial Automation, Numerical Methods
Study Abroad: TECNUN, Universidad de Navarra, San Sebastián, Spain | Spring 2026

EXPERIENCE

Research Assistant, The Pennsylvania State University | Aug-Dec 2025; extended independently 2026 | study & data

- Built the Python dynamic-scheduler model, run visualizations, and report for Prof. Steven Landry's study comparing traffic signal control with reservation-based scheduling at urban intersections.
- Extended the study independently in 2026: rebuilt it as a from-scratch Python microsimulation of a six-signal grid with pedestrians as agents, then tested fixed-time, semi-actuated, and reservation control at five demand levels, eight seeds each (120 runs, 241,938 vehicles).
- Found that reservation scheduling cut stops per vehicle 4.5 to 5.1 times, but under heavy demand raised mean pedestrian wait from 14.3 s to 34.6 s while signal control held near 13 s. The original study could not show this because it did not model pedestrians.
- Added an intersection occupancy check that exposed a safety gap in the initially favored reservation controller.

UX Research & Data Analyst Intern, Glance (InMobi Group) | Mar-May 2026

- Led a four-week diary study of how a four-person household used Glance AI's ambient screens on DirecTV, directing a remote on-site observer through daily logs, photos, and voice notes.
- Compared how trust in and engagement with AI recommendations differed by age and tech comfort, and delivered a synthesis report with a ranking of content performance and specific product recommendations to the Glance team.

RELEVANT PROJECTS

Citi Bike Rebalance & Refill Concept Prototype | React · Leaflet · 2026 | live prototype

- Prototyped a rider map and an operator dispatch view so a rider knows when an empty station will be refilled, instead of walking to it and guessing.
- Classified each station as surplus or deficit from its capacity, target inventory, and current bikes; that status drives the rider's "refill incoming" card and the operator's load/unload stop list, with staffed valet stations excluded from truck routing.
- Truck routes and quantities currently run on a scripted demo scenario; a demand forecast and routing solver on Citi Bike trip data are the planned next phase.

Industrial Automation Team Project, TECNUN | TwinCAT 3 · GRAFCET/GEMMA · Spring 2026 | simulation

- Wrote the state-transition logic for a team-built Beckhoff PLC controller for an airport emergency door with alarm signaling, designed in GRAFCET/GEMMA and implemented in TwinCAT 3.

SKILLS

Modeling & Analysis: Python; MATLAB; Excel; time-stepped and Monte Carlo simulation; Poisson and origin-destination demand; p95 tail analysis
Operations: capacity and flow analysis; inventory and replenishment; routing and scheduling; scenario comparison
Engineering Tools: SolidWorks; Creo; TwinCAT 3 (PLC); React; Git/GitHub; Playwright
Languages: English; Spanish (bilingual)